

Continuum action: evidence and limits

Signal Space / GROSS Test 5

Technical completion: completed; bounded classification: pass.

Run: run-d3333c7602af3e47; analysis: analysis-0001-4dbb966f.

Solver commit: b79f76596450a0b11e03a278f76225a854c75b4f; renderer commit: b79f76596450a0b11e03a278f76225a854c75b4f

200 seeded invertible coframes with nonzero core/neutral gradients and clock fields; six real matter fields, 20 directions and two roots each. Natural units $c=\hbar=m=M_0=1$. The Hermitian coframe and four-dimensional domain are assumptions of SS OCF 1.

Matter Hessian residual $7.43049\text{e-}16$; equations $4.88386\text{e-}16$; characteristic residual $1.76634\text{e-}15$. Gravity: flat linearized harmonic-gauge TT symbol/gauge/Hamiltonian/momentum constraint residual $3.43045\text{e-}16$. Nonlinear Einstein constraints and evolution were not evaluated.

Negative Z and separately named quartic orientation actions are failure controls. Characteristic agreement follows the chosen universal metric coupling; it is an implementation audit, not empirical support for emergence.

No clock, detector record, boundary/continuum convergence, stress-energy conservation in evolution, or nonlinear gravity solution is claimed.

Next: Test 6 bound-clock radial profile/eigenmode screen in the flat decoupling limit. Compare branches and trapping removed before long reception protocols.

characteristic-surfaces

Question: Do all six matter sectors share one coframe null cone in shifted coordinates?

Reading: The six saved roots coincide across 20 directions on 200 coframes; maximum null-polynomial/root residual $1.77\text{e-}15$. Coordinate drift differs across coframes.

Significance: The action implementation retains the selected common matter principal cone even in shifted coordinate frames.

Limitation: Only four coframes and two of six fields are displayed; all saved sectors enter the check. Local frozen coefficients do not establish a metric derived from microscopic reception.

action-audit

Question: Do action derivatives agree with the written equations on nonzero jets?

Reading: Maximum Hessian residual $7.43\text{e-}16$; equation residual $4.88\text{e-}16$; operator metric residual $8.88\text{e-}16$ against $1\text{e-}10$.

Significance: Independent action polarization and complex-step force distinguish the written equations from a copied principal matrix.

Limitation: Finite smooth local jets; no on-shell solution, spatial grid, long-time conservation test or nonlinear gravity constraint solution.

failure-controls

Question: Do unhealthy Z and quartic orientation change distinct accepted properties?

Reading: Negative $Z=-0.5$ fails kinetic positivity; separately versioned quartic orientation action gives speed 0.934947 versus metric speed 1.

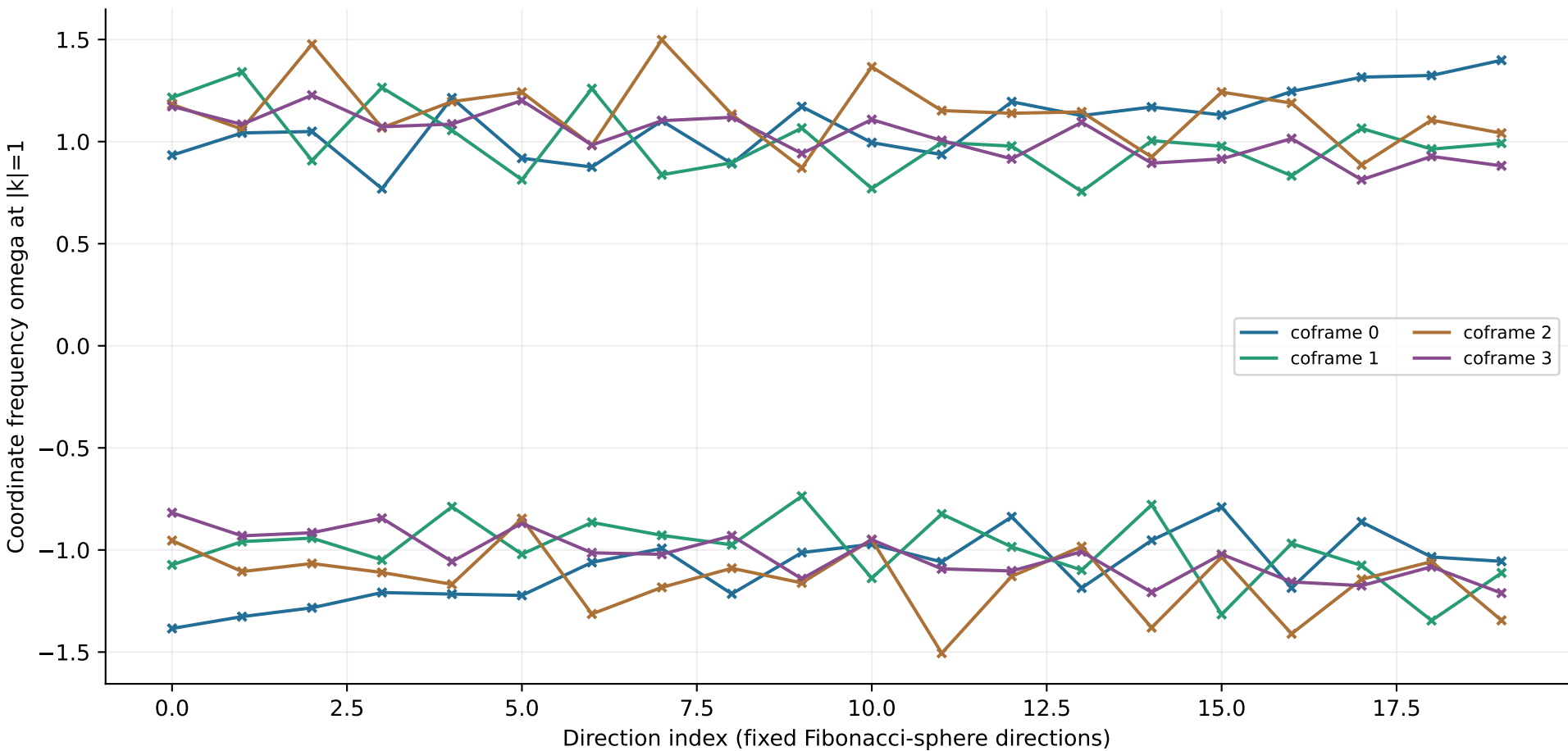
Significance: The audit can detect both a bad kinetic sign and an extra material characteristic that

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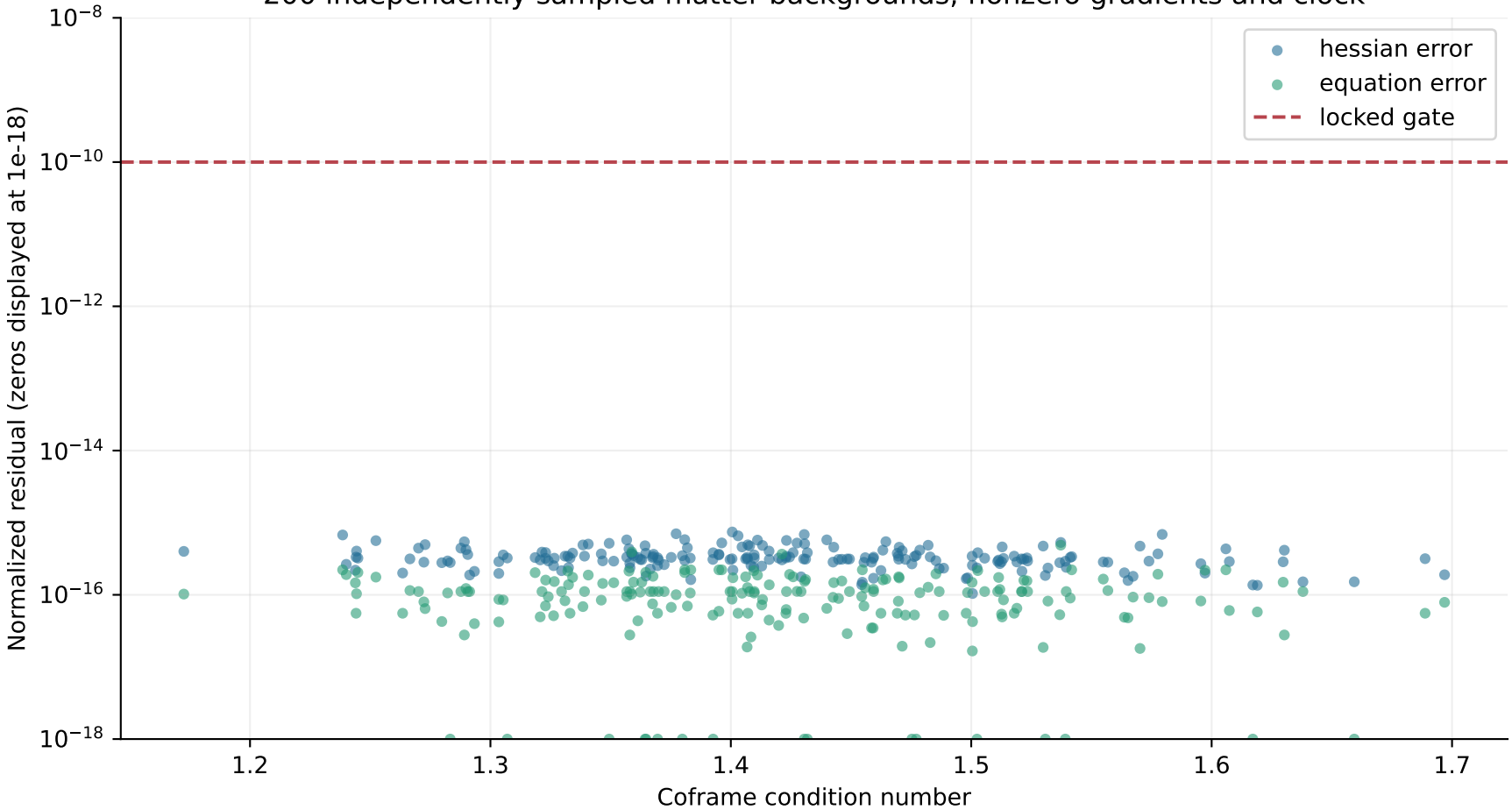
a common coordinate shift cannot remove.

Limitation: The quartic term is excluded from SS OCF 1 and tested on one flat static gradient. It neither proves topology protection nor exhausts all constitutive alternatives.

Lines: core; crosses: neutral field; all six roots checked in saved data

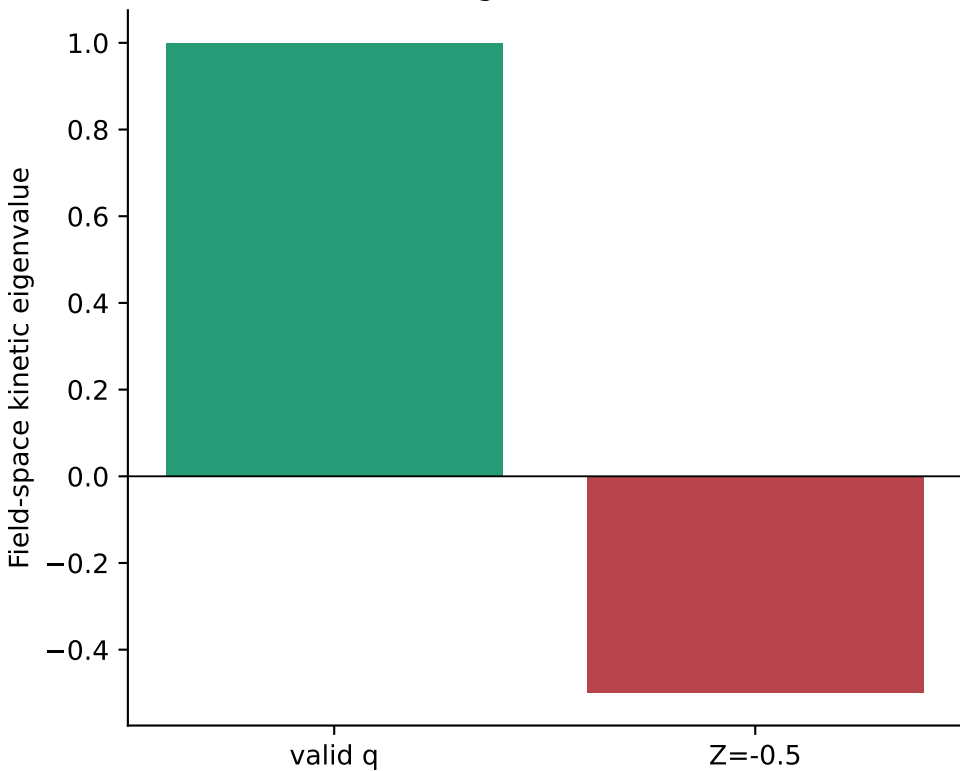


200 independently sampled matter backgrounds; nonzero gradients and clock



Test 5 | Deliberately broken controls, not alternative fits

Artificial negative Z fails health



Separate model: $p=0.6$, $\kappa=0.4$

